

ECEN-738 Project Final Report (Six Phase interleaved Buck Converter)

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Project Video: <https://youtu.be/8RTbGhedBHo>

1. Introduction

This project focuses on the design and simulation of a 6 phase interleaved buck converter that aims to regulate a high current and low voltage output that is suitable for digital processors (SoCs). So, the system is designed to step down a 12 V input that has a load current ranging from 5A to 150A to a regulated 1.1- 0.9 V output voltage with low ripple(100mV) and stable voltage regulation output. The multiphase architecture was chosen to distribute the current across six parallel phases in order to reduce the transient current spikes and ripple. Thus, each phase is interleaved with a 60 degree phase shift which increases the effective switching frequency, reduces the output ripple and improves the transient response by allowing faster current delivery to the system's load. Furthermore, the system is made up of four main stages in a closed loop. It starts with the power stage that consists of MOSFET bridges and inductors. This stage is controlled by gate drivers that apply phase shifted switching signals to each phase. Afterwards, the output is sensed and fed into a feedback non linear control loop that calculates the needed dynamic duty cycle. Lastly, there is the PWM generation stage that converts the duty cycle value into the switching signals that drive the buck converter. The design was implemented in MATLAB Simulink and PLECS and tested under multiple conditions ranging from light load(5A), half load(75A), and full load(150A) in order to evaluate key performance metrics like output voltage regulation, transient response, current sharing between phases, and output ripple. Overall, the design demonstrates stable operation with minimal oscillations.

2. Analysis

The performance of the buck converter is heavily dependent on the selection of the passive components that determine the desired design trade offs between the output ripple, transient response, system output regulation, and maintaining a consistent average current output. The initial components values were estimated using an ideal six phase buck converter equation. But after running the simulation they did not satisfy the dynamic load requirements with the time constraints of the target SoC application due to the interleaving phase frequency. So, the inductance per phase was adjusted to 611 nH which reduces the phase current ripple and provides a stable transient response that the converter could meet (detailed in hardware optimization section of design). Also a 100 nH inductor was added on the shared output to further filter out the combined phases without slowing down the response time. Additionally, a 19 μ F capacitor is used to smooth out the output voltage by filtering out the remaining ripple frequency from the interleaved phases. In order to create the 60 degree phase shift a propagation delay of 555 ns was added on each branch which corresponds to a 300kHz switching frequency per phase. These design choices create a stable current and voltage output with little ripple and a controlled transient response under varying load conditions.

System Dynamics (Natural Second Order System Dynamics)

Upon deriving the small signal model of the control to output transfer function of the 6 phase multiconverter, we get the following transfer function:

$$G_{vd}(s) = V_{in} / (s^2 L_{eq} C_{out} + s (L_{eq} / R_{load}) + 1), \text{ Where,}$$

$$C_{out} = 15\text{mF (updated design parameters)}$$

$$R_{load} = 0.006106 \Omega \parallel 0.01289 \Omega \parallel 0.2 \Omega$$

The damping ratio of the system is given by:

$$\zeta = (1/(2 * R_{load})) * (\text{sqrt}(L_{eq}/C_{out}))$$

Replacing the values and changing R_{load} for different loads, we get the following ζ values:

1. $R_{load} = 0.2 \Rightarrow \zeta = 0.00204$ (Underdamped)
2. $R_{load} = 0.01289 \Rightarrow \zeta = 0.03167$ (Underdamped)
3. $R_{load} = 0.006106 \Rightarrow \zeta = 0.06686$ (Underdamped)

Therefore, the system response for all the three load states is underdamped (this is further corroborated by the responses we see in the simulations section). Whenever the AVP system feeds forward the required duty cycle, we see an underdamped response for all the load step simulations.

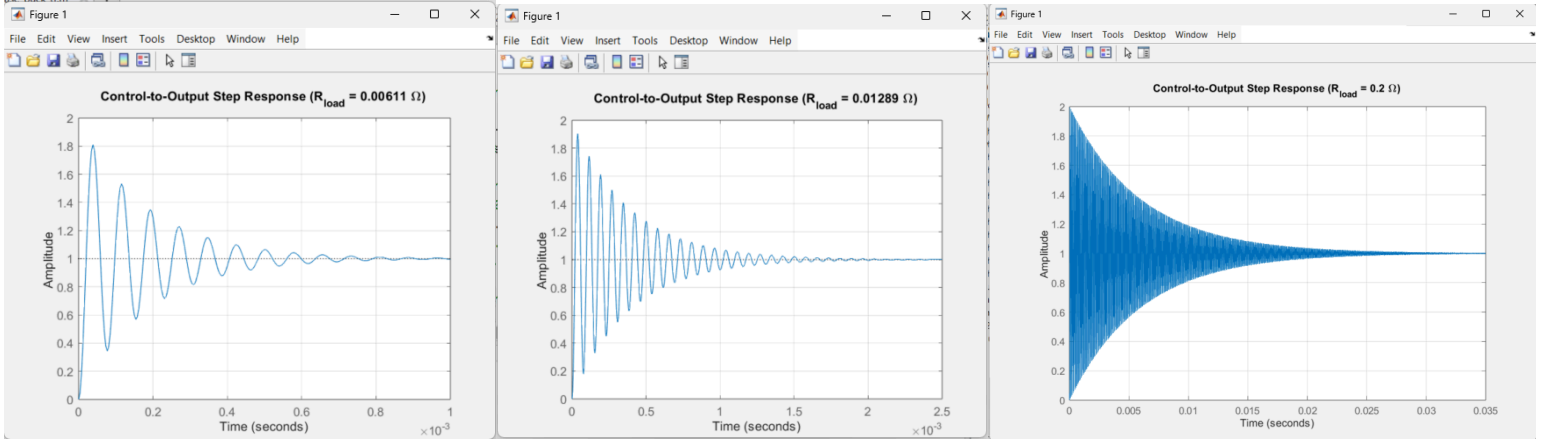


Figure 1 - PLECs simulation step response for resistive loads 6.11 (150A), 12.89 (75A), and 200 (5A) mΩ

3. Design

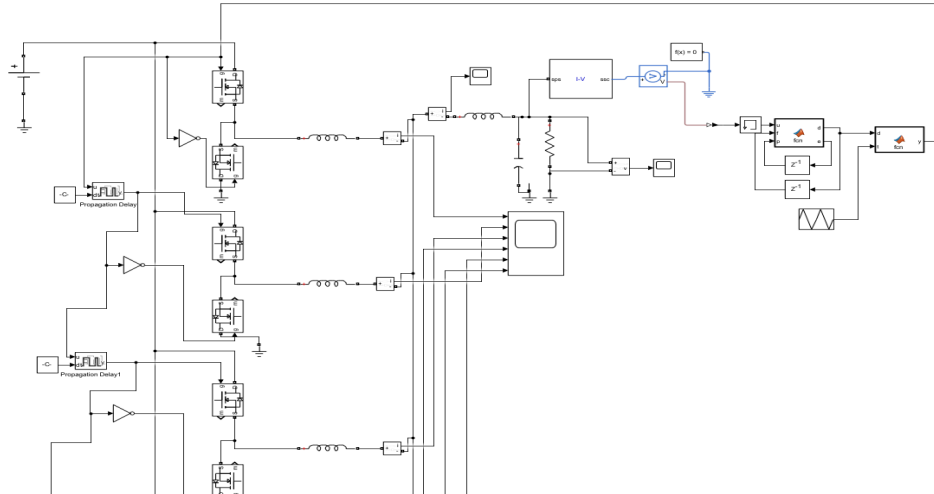


Figure 2 - The 6-phase buck converter MATLAB design (first three phases shown)

Figure 2 above depicts the majority of the converter design. Each of the six phases is defined by a high and low-side FET which are both driven by the feedback network. These FETs enable the flow of current (or lack thereof) through a 611 nH inductor for each phase, which are all connected to contribute towards the output current at the load.

The feedback network seen below in Figure 3 is split into two main parts, both implemented using MATLAB function blocks: the duty cycle generator and the PWM signal generator. The duty cycle generator takes the form of a function with three inputs (current converter voltage output U , previous duty cycle output F , and previous error P) and two outputs (duty cycle D and error E). Based on the value of the current converter output voltage relative to the desired output of 0.9 V, the duty cycle will be modified according to the PI controller formula:

$$D = F \pm (K_1 * |E|) \pm (K_2 * |P|)$$

Here, K_1 and K_2 are constants which were determined via trial and error. In the final design, $K_1 = 0.025$ and $K_2 = 0.017$. From here, the calculated duty cycle D and a 300 kHz triangle wave T are fed as inputs to the second function block: the PWM signal generator. This block converts the duty cycle (between 0 and 1) into a simulated “reference voltage” (between -1 and 1) with the formula:

$$V_{ref} = (2(1 - D)) - 1$$

This reference voltage value is then used alongside the current triangle waveform value like a comparator to produce the PWM signal Y , where the output is 1V when $T > V_{ref}$ and -1V otherwise. Lastly, this signal is used to drive the FETs at each of the six stages of the converter. A propagation delay block with an incrementing value of 555 ns is used to interface between the PWM generator and each phase to produce an appropriate delay. The delayed signal directly drives the high-side FET while low-side FET receives the NOT of the signal.

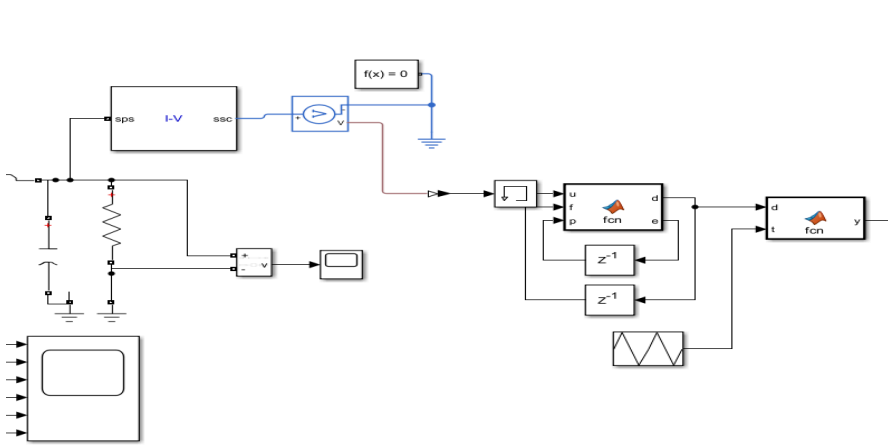


Figure 3 - Converter feedback network

To further improve the system operation and functionality, apart from the traditional PI control which is linear, a non-linear approach was selected.

Transient Optimization & Hybrid Control Implementation

1. **Hardware Optimization:** Increased output capacitance (15mF) and reduced phase inductance (60nH) to minimize voltage ripple and significantly accelerate current response during rapid load steps. This pushed the converter from CCM to FCCM (Forced Continuous Conduction Mode).
2. **Active Voltage Positioning (AVP):** Replaced standard PI regulation with AVP to dynamically adjust the steady-state voltage reference based on load requirements. This creates essential transient headroom to manage massive 5A–150A load steps within the strict $1V \pm 100mV$ SoC threshold, eliminating the need for extra bulky capacitor banks.
3. **Non-Linear Transient Control (AOT/COT):** Because AVP is too slow to detect immediate voltage dips, an Adaptive/Constant On-Time (AOT/COT) non-linear control was implemented. This instantly overrides AVP during sudden load steps to maintain strict output regulation, seamlessly handing control back to AVP once steady-state is reached.

AVP Logic

Replaces rigid, standard voltage regulation by introducing a controlled voltage droop proportional to the load current. This optimizes efficiency and stability during massive, instantaneous load steps (e.g., 5A to 150A). The characteristic output impedance of the converter is given by:

$$Z_o = \sqrt{L_{eq}/C}$$

$$Z_o = \sqrt{10e - 9/15e - 3}$$

$$Z_o = 0.815m\Omega$$

Using the characteristic output impedance, we can calculate the max voltage step that happens during a max load change (since there are only 3 load states, the maximum load current change is 145A (5A-150A)).

$$\Delta V = \Delta I * Z_o$$

$$\Delta V = 145 A * 0.815m\Omega$$

$$\Delta V = 118mV$$

Sudden load changes cause unavoidable ~120mV voltage spikes or dips. A static 1V reference would cause these $\pm 120mV$ transients to breach the SoC's strict 0.9V–1.1V safe operating window.

AVP dynamically shifts the reference voltage based on load. At high currents, the reference is intentionally lowered. If the load suddenly drops, the resulting ~120mV voltage spike is safely absorbed and contained below the 1.1V maximum threshold.

AVP is driven by the equation:

$$V_{ref_{act}} = V_{ref_{opt}} - V_{droop}$$

Where,

$V_{ref_{opt}}$ = Optimal reference voltage (This is also the reference voltage to be followed at no-load condition)

$V_{ref_{act}}$ = Actual reference voltage from which the Duty Cycle D will be calculated

V_{droop} = Droop voltage (120mV in our system due to maximum output characteristic impedance)

Note: $V_{droop} = R_{droop} * I_{load}$

$$120mV = R_{droop} * 150A$$

$$R_{droop} = 0.8m\Omega$$

Assuming the $V_{ref_{act}} = 0.9V$ is the voltage output at the highest load demand of 150A (we assume such a situation of the lowest voltage threshold because 150A is the highest load being supplied to the SoC, any load demand after that would only be load decrement, which would only cause a voltage spike upwards),

$$0.9V = V_{ref_{act}} - 120mV$$

$$V_{ref_{act}} = 1.02V$$

Considering capacitor ripple and a little headroom for the system,

$$V_{ref_{act}} = 1.05V$$

This implies, by setting the system to follow a dynamic voltage reference, we can allow the natural response of the system to settle on its own without additional linear control application.

AOT/COT Logic

1. **Operating Point Shift:** Steady-state phase-shifted switching is too slow for rapid load jumps. Non-linear AOT/COT control temporarily overrides standard switching by forcing all phases ON simultaneously. This immediately pumps inductor current to the load, preventing capacitor drain and keeping the voltage regulated.
2. **Trigger Mechanism:** The logic focuses strictly on low-to-high load transitions. If the output voltage drops to the 0.9V lower limit, the controller senses the deficit and fires a Constant On-Time (COT) pulse to rapidly shift the operating current point.
3. **Control Handoff:** Once the inductors are carrying the required current, the non-linear control seamlessly hands operation back to the AVP mechanism to maintain steady-state voltage thresholds.
4. **Pulse Calibration:** The system manages three load states (5A, 75A, 150A). The COT pulse duration is mathematically calibrated for a ~75A jump. A single pulse satisfies a 75A step, while the controller fires twice consecutively to bridge a massive 5A to 150A step.
5. **High-to-Low Transitions:** Handled automatically by natural system dynamics. Draining current briefly charges the output capacitor, creating a controlled voltage spike that safely stays within rated limits due to AVP positioning and optimized hardware.

The ON time (assuming a 75A jump) of the pulse can be calculated as follows:

Note: Since the state jumps of the load go from 5A-75A-150A, a max current jump of 75A is the limit. 75A is almost symmetrically distributed i.e. 5A- 75A is a 70A jump while a 5A to 150A is a 145A (approx 75*2= 150A) jump and a 75A- 150A jump is a 75A jump. For higher jumps the COT pulse triggers twice to meet the deficit while for the smaller jumps a single pulse is sufficient. The excess current is drained off the system naturally according to the system dynamics.

$$V_L = L_{eq} * di/dt$$

$$V_L = V_{in} - V_{out} = 12 - 1.05 = 10.95V$$

$$10.95 = L_{eq} * \Delta I / \Delta t$$

$$L_{eq} = 60e-9/6 = 10e-9 \text{ H (when all phases are turned ON, the 6 inductors are parallel to each other)}$$

$$\Delta I = 75A$$

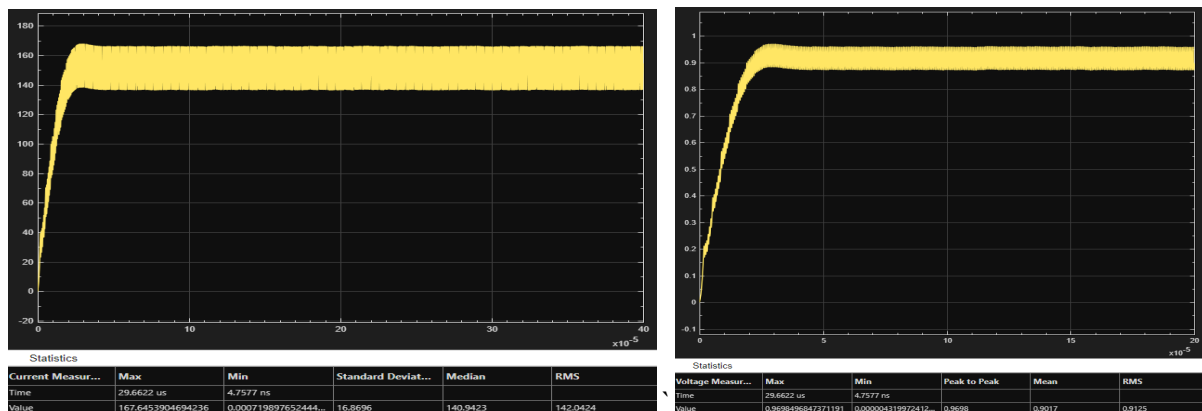
Substituting the values, we get

$$\Delta t = 68.5\text{ns}$$

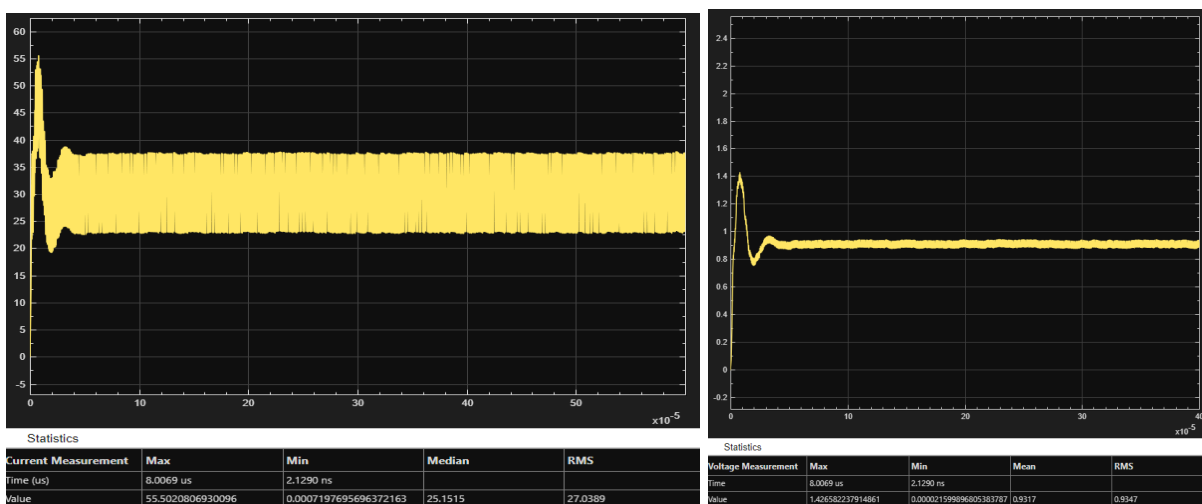
Therefore, we can fix the ON time for the COT control as 68.5ns.

4. Simulations and Discussion of Results

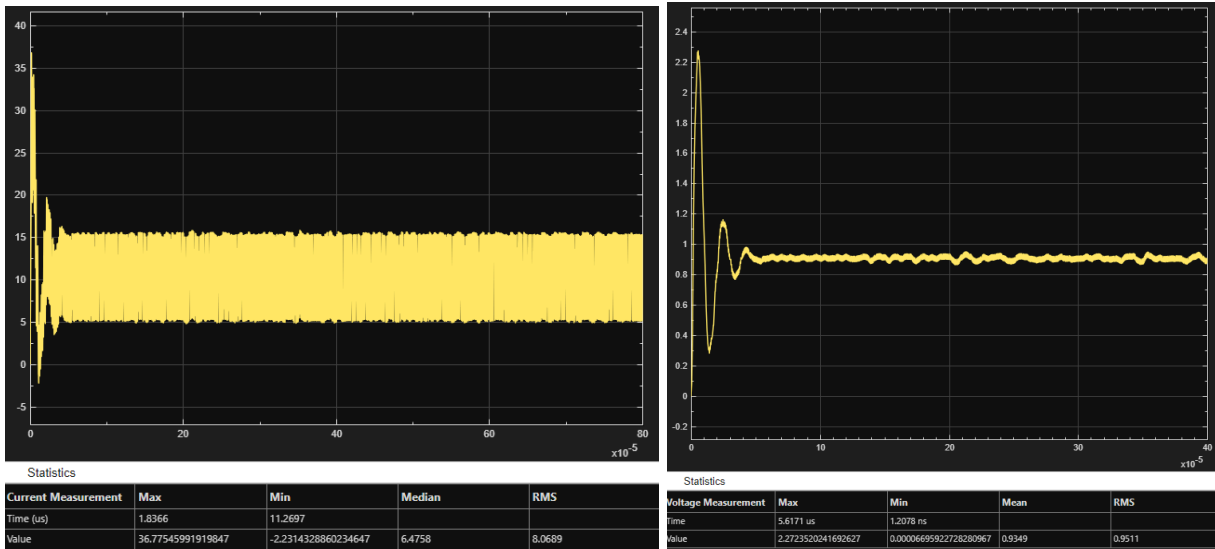
Figures 4 through 11 below show the load voltage and current waveforms for a series of resistive loads which elicit the range of current demands for which this converter was designed, which is 5 A to 150 A. With a steady output voltage of 0.9 V, this results in a range of $0.9 / 5 = 180\text{ m}\Omega$ to $0.9 / 150 = 6\text{ m}\Omega$. To support a higher current, the PI controller was tuned to provide the best performance to loads at the smaller end of this range. As a result, Figures 3 and 4 show waveforms with minimal overshoot which settle very closely to the expected voltage and current values of the design around 0.9 V and 150 A. As the load increases in Figures 5 through 10, the overshoot also appears to increase linearly. However, the feedback network ensures that the RMS values of the output settle at their expected values after 5 to 20 μsec for both voltage and current.



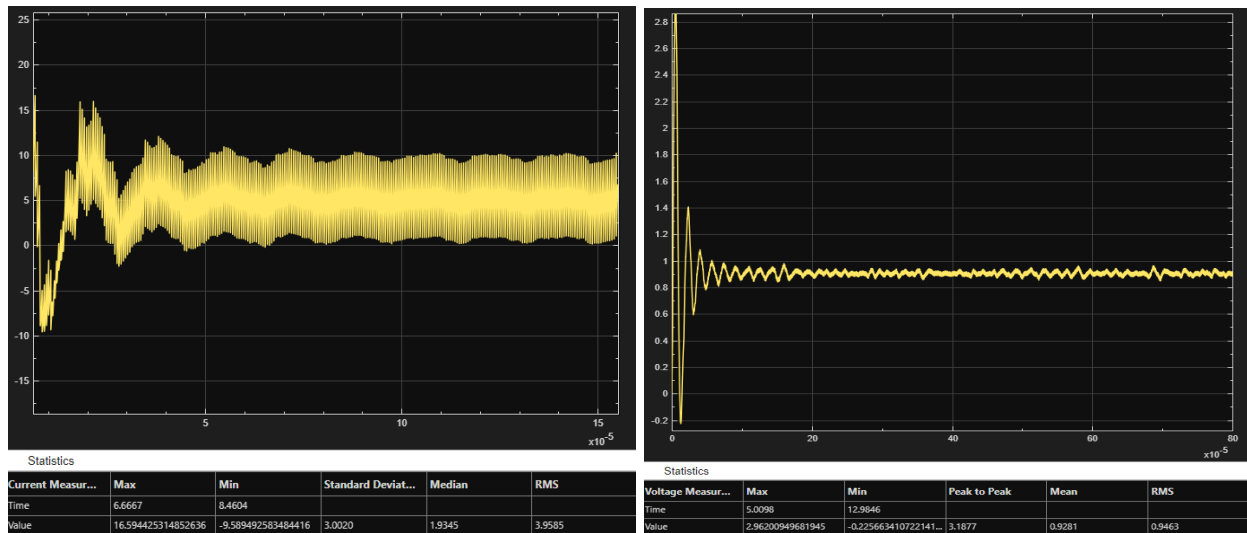
Figures 4 & 5 - Output current (left) and voltage (right) response for a 6 m Ω load



Figures 6 & 7 - Output current (left) and voltage (right) response for a 30 m Ω load



Figures 8 & 9 - Output current (left) and voltage (right) response for a 93 mΩ load



Figures 10 & 11 - Output current (left) and voltage (right) response for a 180 mΩ load

Figure 12 below shows the increasing delay between each phase of the output. This can be approximated by the time at which each current becomes ≥ 0 A. The time between peaks of the current ripple, i.e. the period of the waveform, was measured to be 3.358 μsec .

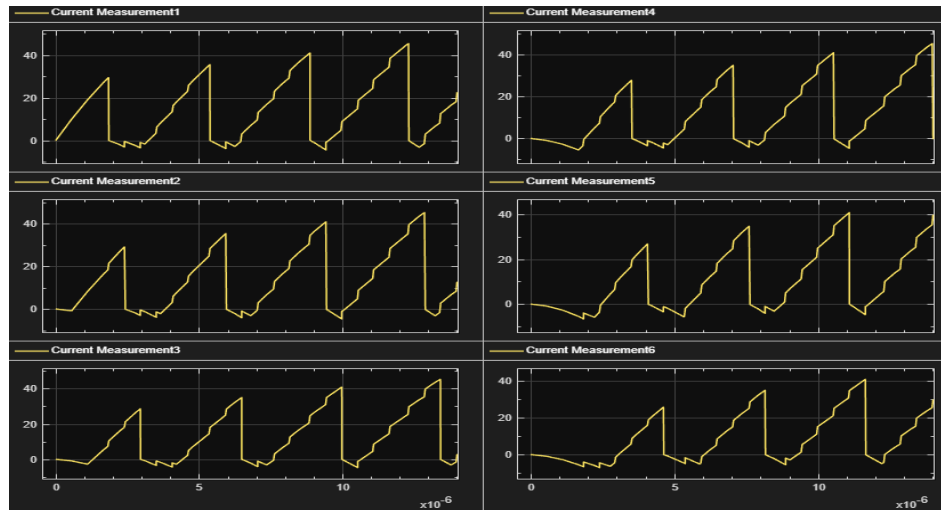


Figure 12 - Current response by phase for a 6 mΩ load

Table 1 - Output current phase delay for a 6 mΩ load

Phase	Time Current ≥ 0 A (μsec)	Phase Delay (deg)
1	0.000	0
2	0.613	65.718
3	1.301	139.476
4	1.879	201.441
5	2.468	264.586
6	3.058	327.838

In Table 1 above, the phase delay is calculated as $360^\circ * (\text{time} / 3.358)$. The RMS average difference in delay between phases comes out to 61.402° , which is expected since $360^\circ / 6 \text{ phases} = 60^\circ$. This indicates that the timing between phases works as expected.

With regards to the PLECS simulation, the linear control using the PI block manages to reach the required output voltage and current needs but suffers through intense transients and operating point issues. Circumventing this problem, the non-linear AVP coupled COT control operates the system much quicker and reaches the required load requirements at a much faster and reliable pace without compromising on robustness and Safe Operating Region (SOR).

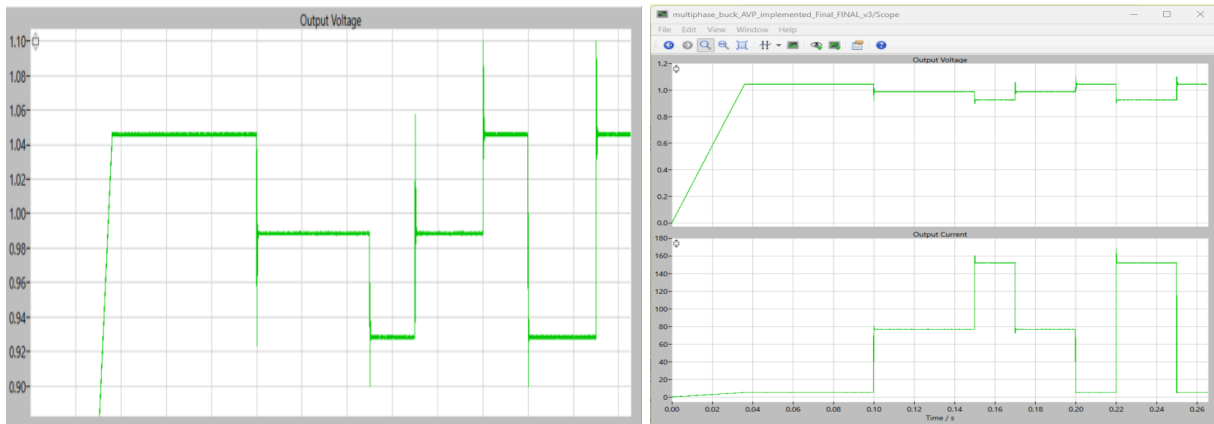


Figure 13: Output voltage and load current for different load steps with Soft- Start, AVP and COT control implemented

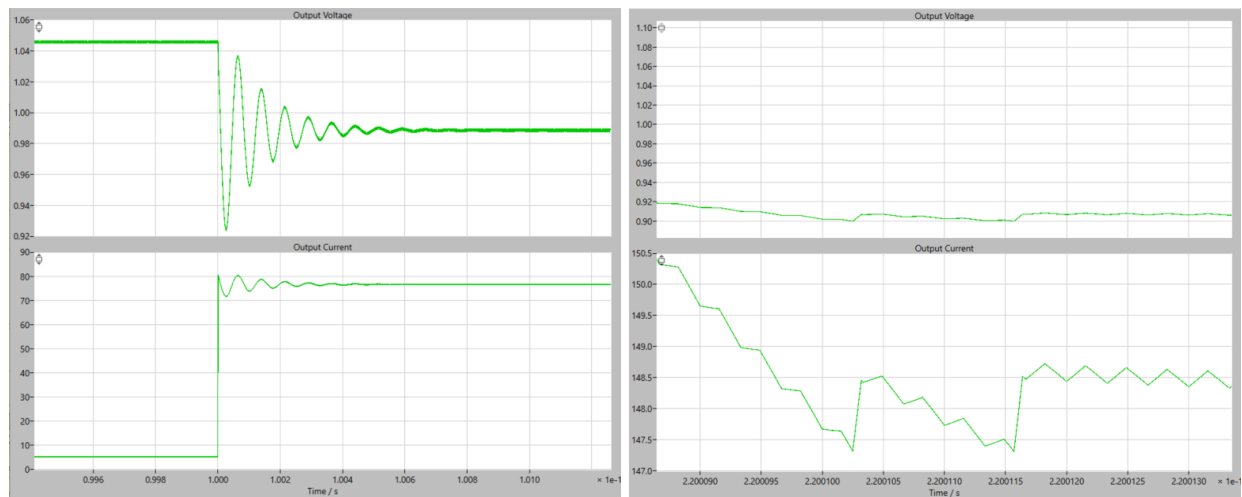


Figure 14: COT 0.9V tripwire and Pulse Operation (2 pulses) (right) Natural system response for 5A-75A load step (left)

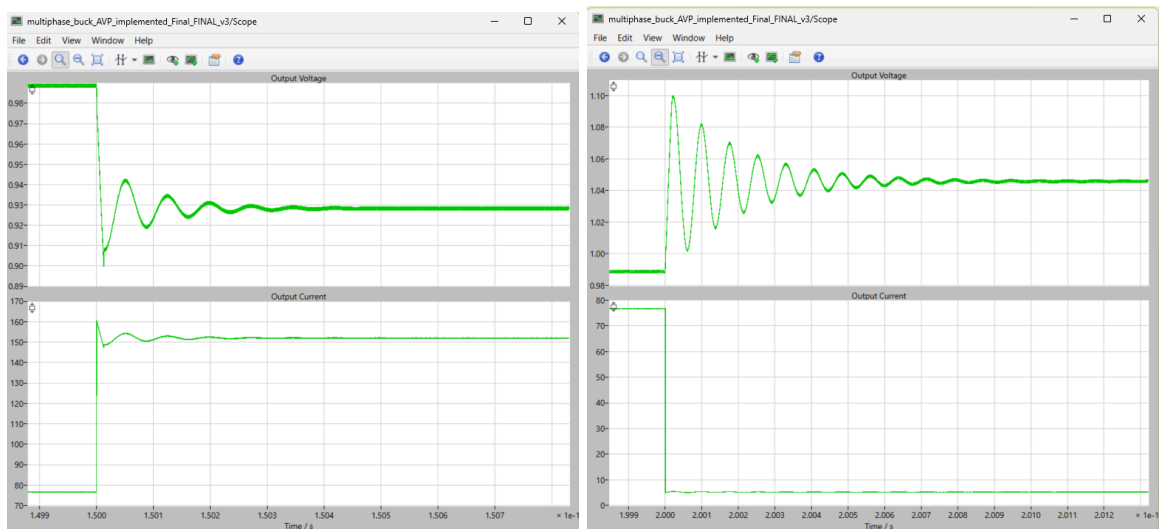


Figure 15: 75A-150A load step (left) 75A- 5A load step (right)

5. Conclusion

The MATLAB design successfully demonstrated coordinated six-phase operation, delivering the required output voltage and current with reliable startup regulation in under 100 μ s. This validated the strength of the underlying architecture and showed that the converter could meet its performance targets under nominal conditions. However, the design also revealed consistent overshoot during startup, high ripples, particularly at lower load currents, highlighting the limitations of relying solely on a linear PI controller for a system with rapidly changing demands.

Because the SoC load exhibits highly nonlinear, fast transient behavior, the PI controller was unable to maintain optimal performance across the full operating range. To address these shortcomings, a nonlinear control strategy (simulated in PLECS) was developed to better handle steep load transitions and reduce overshoot. This approach directly targets the dynamic nature of the SoC load and resolves the weaknesses observed in the PI-based control logic, resulting in a more robust and responsive converter design.

6. References

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- [2] Peterson, Zach. "Multiphase Buck Converter: The Fundamentals." *YouTube*, Altium Academy, 22 June 2023, www.youtube.com/watch?v=_qnU9niqlb4.
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